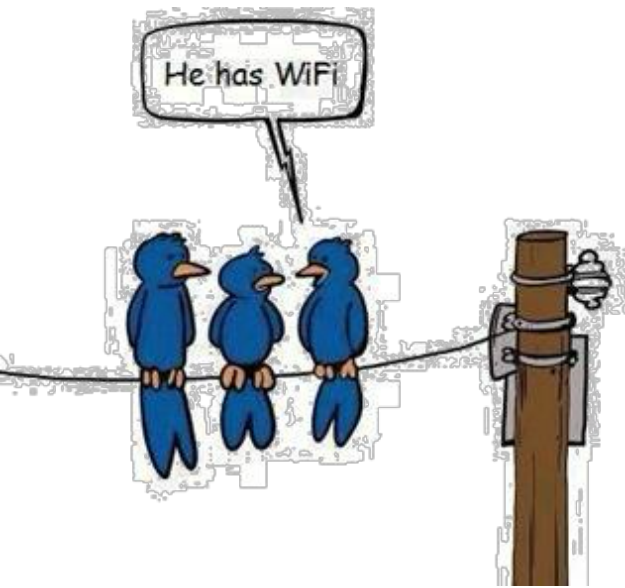


ECSE 4560/ 6560

Modern Communication Systems

Lecture 19: MIMO Part4: Multiplexing



Ish Jain, Fall 2025



Announcements

- Join Piazza: <https://piazza.com/rpi/fall2025/ecse45606560>
- Per-Class Quiz
 - Posted on LMS; Due Today 11:59 PM
- Attendance Link: <https://forms.office.com/r/p6P9YrN1d8>
 - Today's Magic Word: **Multiplexing**
- Projects
 - Group formation [MCS Fall25 project.xlsx](#)
 - Project mid-term report due ~~Friday 11/7~~ **(Extension?)** *Monday*
 - Final Project Presentation and Demos: Dec 9 and Dec 12
 - Project Report and peer evaluation: Finals exam week
- HW/Exams Dates
 - HW5 (MIMO Theory) due **Mon 11/10**. HW6 (MIMO code) due 11/14
 - Exam 2: **Friday Nov 21**

Extra Reading on Piazza

- Principles of Modern Wireless Communication Systems Theory and Practice; by Aditya K. Jagannatham, IIT Kanpur Book
 - https://piazza.com/class_profile/get_resource/mevvk9kirhg2fu/mgb9esiush64ae
 - Parts of Chapter 1,2,3,4,7
 - **MIMO Chapter 6**

Today's Class

- MIMO Gains

- Receiver Diversity
- Transmitter Diversity
- Space-Time codes (Alamouti Codes)

CSIT

3 diversity

NO CSIT

- MIMO Multiplexing Gains

- Zero-forcing
- SVD
- MU MIMO

OFDM

OFDM

- Clock Synchronization in MIMO

^

MIMO Gains

$$C = B \log(1 + \text{SNR})$$

capacity or rate
bits/sec

Multiplexing Gain:

- Send multiple packets at the same time
- $N \times N$ MIMO $\rightarrow \times N$ more packets
- Data Rate: $\propto N \log(\text{SNR}/N)$

$N \times M$
 $N > 1$
 $M > 1$
multiplexing gain

Diversity Gain:

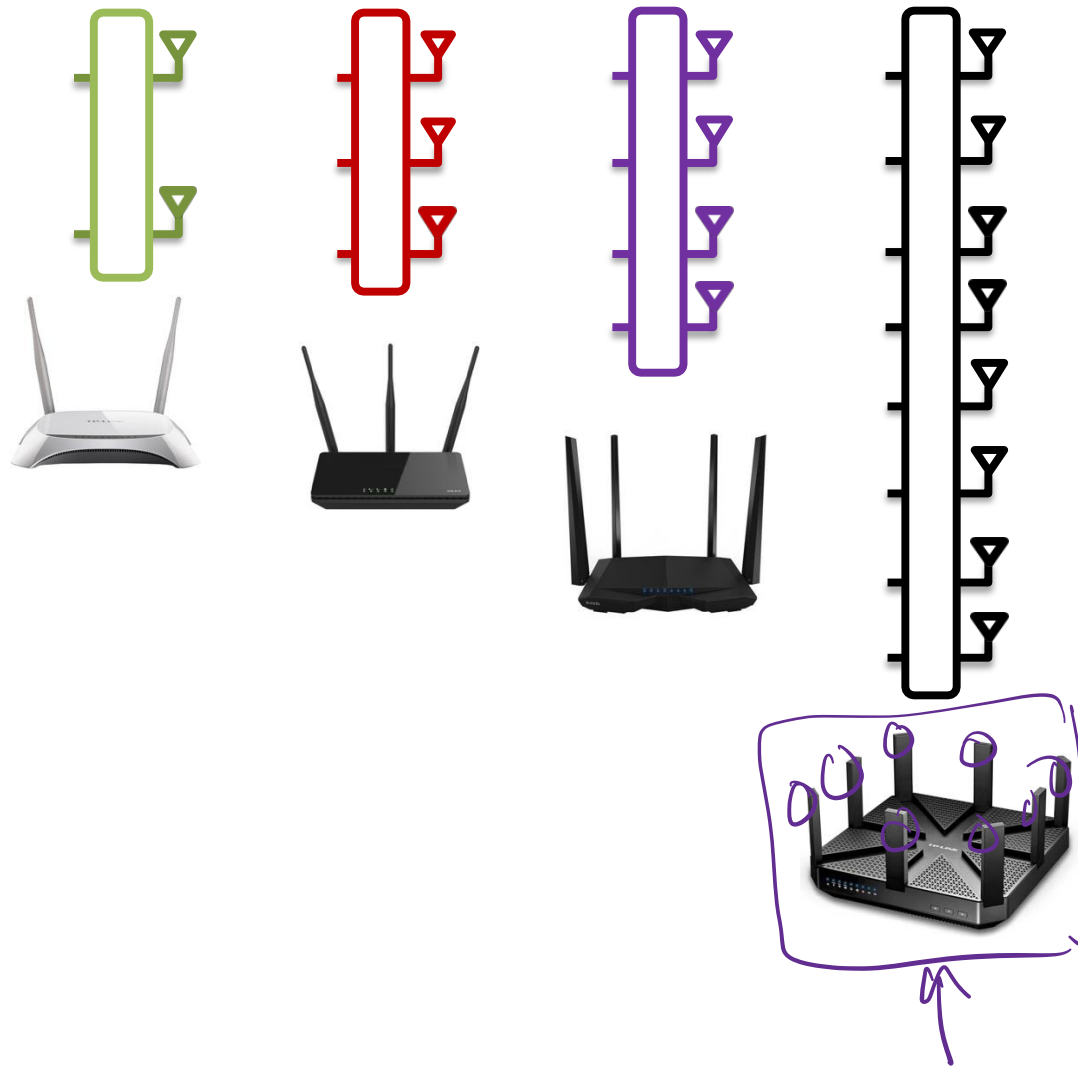
- Increase SNR of the received packets
- $N \times 1$ or $1 \times N$ MIMO $\rightarrow \times \log N$ rate
- $N \times N$ MIMO $\rightarrow \times \log N^2$ data rate
- Data Rate $\propto \log(\text{SNR} \times N^2)$

$$\left(|h_{11}|^2 + |h_{12}|^2 + |h_{21}|^2 + |h_{22}|^2 \right) \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix}$$

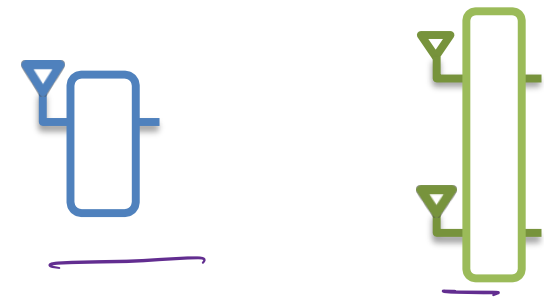
$P/0.2$

MIMO: Multiple TX-RX streams

WiFi Access Points 2 to 8 MIMO



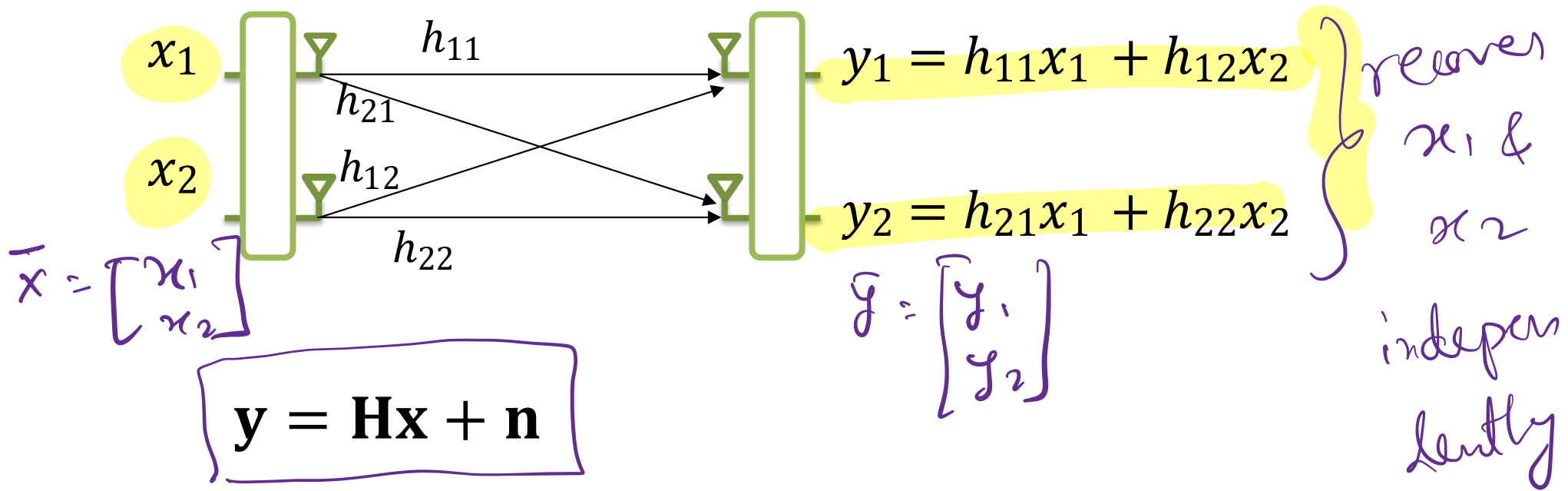
WiFi Devices 1 to 2 MIMO



- Power
- Form Factor & Antenna Separation
- Most mobile phones support 2 MIMO

8×2 MIMO
↙ 2 data streams
↘ 8 antennas for diversity

Zero-forcing MIMO 2x2 example when H is invertible



$$\tilde{\mathbf{x}} = \mathbf{H}^{-1}\mathbf{y} = \mathbf{H}^{-1}\mathbf{H}\mathbf{x} + \mathbf{H}^{-1}\mathbf{n} = \mathbf{x} + \mathbf{H}^{-1}\mathbf{n}$$

$$E(|x_1|^2) = P_1$$

$$E(|n_1|^2) = N_0$$

$$= E(|n_2|^2) = N_0$$

$$SNR = \frac{P_1}{N_0}$$

$$(\mathbf{H}^T \mathbf{H}) = \mathbf{I}_{2 \times 2}$$

$$\frac{E|x_1|^2}{E|\tilde{n}_1|^2} = \frac{P_1}{(\mathbf{H}^T)_{11}^2 N_0}$$

$$\begin{bmatrix} \tilde{n}_1 \\ \tilde{n}_2 \end{bmatrix} = \mathbf{H}^{-1} \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

$$x_1 \sim E |x_1|^2 = P_1$$

$$n_1 \sim \mathcal{CN}(0, N_0)$$

$$n_2 \sim \mathcal{CN}(0, N_0)$$

$$E(|n_1|^2) = \sigma_n^2 = N_0$$

noise
spectral
density

$$x_1 \rightarrow \text{BPSK}(-1, 1)$$

$$\begin{bmatrix} \tilde{z}_1 \\ \tilde{z}_2 \end{bmatrix} = H^{-1} \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

$$\tilde{z}_1 = (H^{-1})_{1,1} n_1 + (H^{-1})_{1,2} n_2$$

$$\begin{aligned} E(|\tilde{z}_1|^2) &= (H^{-1})_{1,1}^2 E(|n_1|^2) \\ &+ (H^{-1})_{1,2}^2 E(|n_2|^2) \\ &= \left[\left((H^{-1})_{1,1} \right)^2 + \left((H^{-1})_{1,2} \right)^2 \right] \sigma_n^2 \end{aligned}$$

Signal power

total power $P = E|z_1|^2 + E|z_2|^2$

$$P_1 = E(|z_1|^2) = E(|z_2|^2)$$

$$P_1 = P_2 = P/2 \quad (\text{val power}) \quad P_2$$

$$\text{capacity} = \log(1 + \text{SNR})$$

$$\text{cap}(x_1) + \text{cap}(x_2) \\ = \text{total capacity} = \text{rate}$$

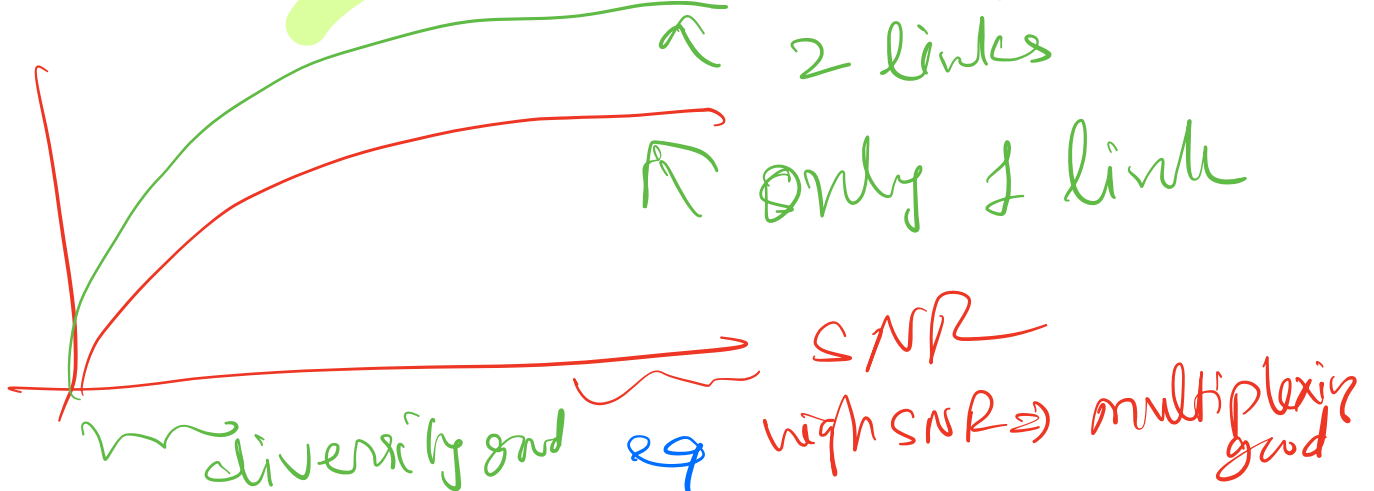
total rate / capacity

$$\propto \log(1 + \text{SNR}_1) + \log(1 + \text{SNR}_2)$$

$$= \log\left(1 + \frac{P_1}{|H|_{1:}^2 N_0}\right) + \text{similar}$$

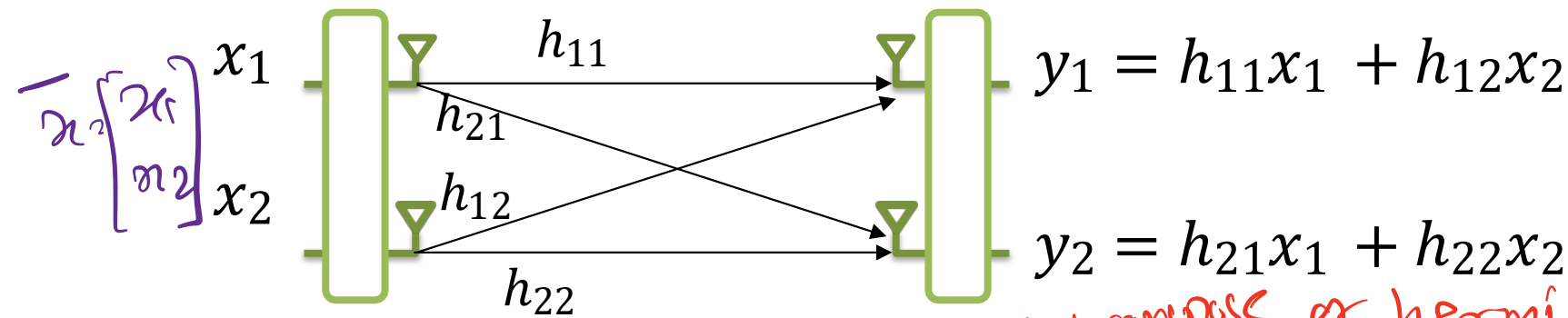
$$= 2 \log\left(1 + \frac{P}{2 |H^*|_{1:}^2 N_0}\right)$$

cap



Zero-forcing based MIMO

H may not be invertible



conj. transpose or hermitian

$$\hat{\mathbf{x}} = (\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H \mathbf{y}$$

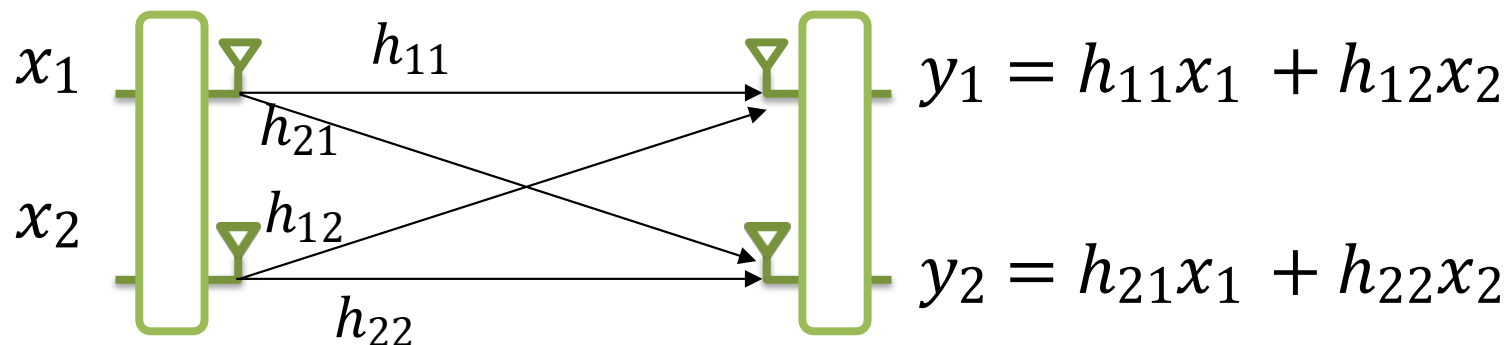
$$\bar{\mathbf{y}} = \mathbf{H} \bar{\mathbf{x}} + \bar{\mathbf{n}}$$

pseudo inverse

$$\bar{\hat{\mathbf{x}}} = (\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H \bar{\mathbf{y}}$$

$$\bar{\hat{\mathbf{x}}} = \cancel{(\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H} \mathbf{H} \bar{\mathbf{x}} + \underbrace{(\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H}_{\text{this may amplify noise}} \bar{\mathbf{n}}$$

Zero-forcing based MIMO



Pros

- Very simple, low latency; no transmitter CSI required.
- Completely removes inter-stream interference when $\mathbf{H}$ is well-conditioned.
- Works symbol-by-symbol; easy to pair with standard demappers.

Cons

- **Noise amplification** when $\mathbf{H}$ is ill-conditioned (small singular values).
- Sensitive to channel-estimation error.
- Not capacity-optimal; BER can be far from MMSE/ML at low-mid SNR.
- Fails if $\mathbf{H}$ is rank-deficient (needs regularization).

Question 3: Simple 2×2 MIMO with Zero-Forcing

Setup. Consider a narrowband flat-fading 2×2 MIMO link for a single channel use:

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}, \quad \mathbf{H} \in \mathbb{C}^{2 \times 2} \text{ invertible,}$$

with $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $\mathbb{E}[|x_1|^2] = \mathbb{E}[|x_2|^2] = P$, and $\mathbf{n} \sim \mathcal{CN}(\mathbf{0}, N_0 \mathbf{I}_2)$. The receiver knows $\mathbf{H}$ perfectly.

- (a) **ZF equalizer and effective noise (one-step derivation).** Use the zero-forcing equalizer $\mathbf{W} = \mathbf{H}^{-1}$. Show that

$$\hat{\mathbf{x}} = \mathbf{W}\mathbf{y} = \mathbf{x} + \mathbf{H}^{-1}\mathbf{n}.$$

Write the two decision statistics explicitly:

$$\hat{x}_1 = [\mathbf{H}^{-1}]_{1,:} \mathbf{y}, \quad \hat{x}_2 = [\mathbf{H}^{-1}]_{2,:} \mathbf{y},$$

and identify the corresponding effective noise terms on each stream.

- (b) **Numeric evaluation.** Let

$$\mathbf{H} = \begin{bmatrix} 1 & 0.5 \\ -0.3 & 1.2 \end{bmatrix}, \quad P = 1, \quad N_0 = 0.2.$$

Compute:

- (i) $\mathbf{H}^{-1}$ (show your intermediate steps briefly).
- (ii) The squared row norms of $\mathbf{H}^{-1}$: $\|\mathbf{w}_1\|^2 = \|[\mathbf{H}^{-1}]_{1,:}\|^2$, $\|\mathbf{w}_2\|^2 = \|[\mathbf{H}^{-1}]_{2,:}\|^2$.
- (iii) The post-ZF SNRs using these norms (no extra matrices):

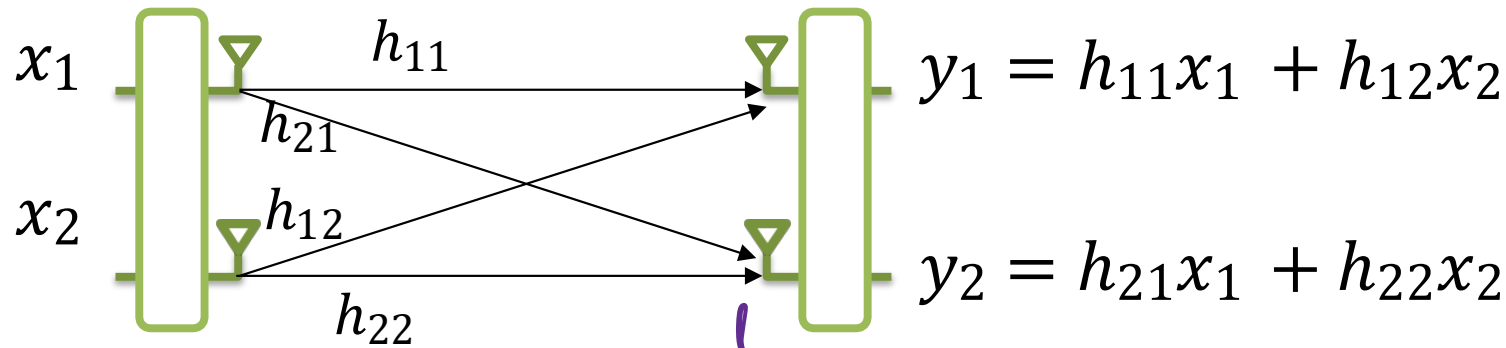
$$\gamma_1 = \frac{P}{N_0 \|\mathbf{w}_1\|^2}, \quad \gamma_2 = \frac{P}{N_0 \|\mathbf{w}_2\|^2}.$$

- (iv) Which stream is more noise-enhanced? State your answer and the SNR ratio γ_1/γ_2 in dB.

SVD-based MIMO

better gain
compared to
ZF

but
requires
CSIT



book
Ch. 6

$$\mathbf{H} = \begin{bmatrix} u_{11} & u_{12} & \dots & u_{1t} \\ u_{21} & u_{22} & \dots & u_{2t} \\ \vdots & \vdots & \ddots & \vdots \\ \underbrace{u_{r1}}_{\mathbf{u}_1} & \underbrace{u_{r2}}_{\mathbf{u}_2} & \dots & \underbrace{u_{rt}}_{\mathbf{u}_t} \end{bmatrix} \begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_t \end{bmatrix}$$

orthonormal

$\mathbf{U}$

Σ

diagonal
real numbers
 σ_1, σ_2
also called
singular
values

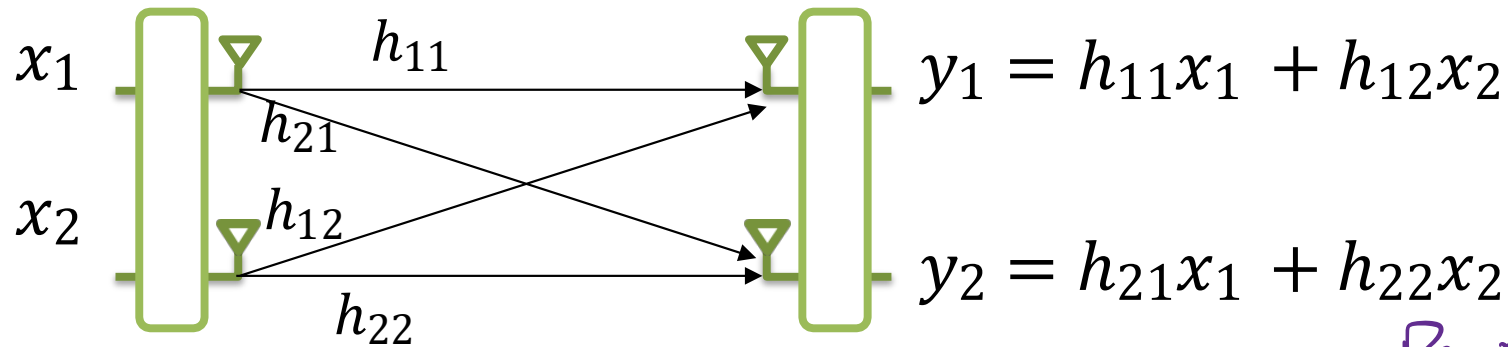
$$\begin{bmatrix} v_{11}^* & v_{21}^* & \dots & v_{t1}^* \\ v_{12}^* & v_{22}^* & \dots & v_{t2}^* \\ \vdots & \vdots & \ddots & \vdots \\ v_{1t}^* & v_{2t}^* & \dots & v_{tt}^* \end{bmatrix} \begin{matrix} \} \mathbf{v}_1^H \\ \} \mathbf{v}_2^H \\ \\ \} \mathbf{v}_t^H \end{matrix}$$

$\mathbf{V}^H$

each row is
orthonormal
vectors

$$\mathbf{H} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H$$

SVD-based MIMO



Properties of SVD

$$\mathbf{H} = \underbrace{\begin{bmatrix} u_{11} & u_{12} & \dots & u_{1t} \\ u_{21} & u_{22} & \dots & u_{2t} \\ \vdots & \vdots & \ddots & \vdots \\ \underbrace{u_{r1}}_{\mathbf{u}_1} & \underbrace{u_{r2}}_{\mathbf{u}_2} & \dots & \underbrace{u_{rt}}_{\mathbf{u}_t} \end{bmatrix}}_{\mathbf{U}} \underbrace{\begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_t \end{bmatrix}}_{\mathbf{\Sigma}}$$

$$\|\mathbf{u}_i\|^2 = \|\mathbf{v}_i\|^2 = 1, 1 \leq i \leq t$$

$$\mathbf{u}_i^H \mathbf{u}_j = \mathbf{v}_i^H \mathbf{v}_j = 0, i \neq j, 1 \leq i, j \leq t$$

$$\mathbf{V}^H \mathbf{V} = \mathbf{V} \mathbf{V}^H = \mathbf{I}_t$$

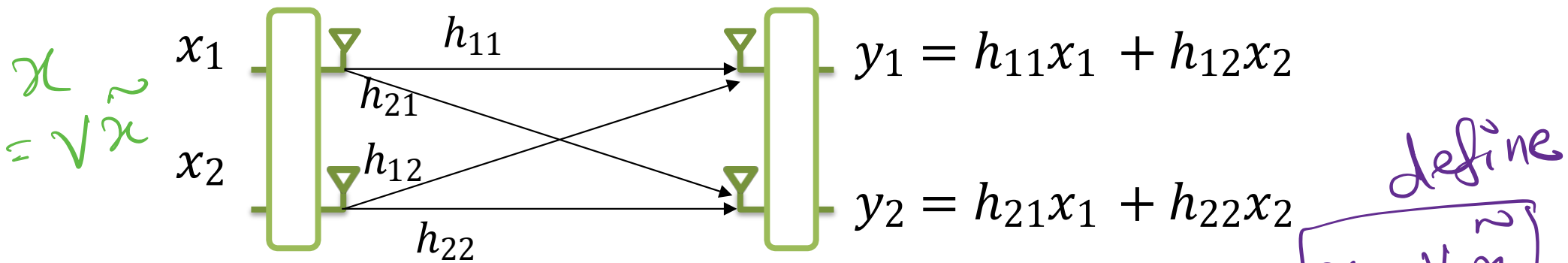
$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_t$$

$$\underbrace{\begin{bmatrix} v_{11}^* & v_{21}^* & \dots & v_{t1}^* \\ v_{12}^* & v_{22}^* & \dots & v_{t2}^* \\ \vdots & \vdots & \ddots & \vdots \\ v_{1t}^* & v_{2t}^* & \dots & v_{tt}^* \end{bmatrix}}_{\mathbf{V}^H},$$

$\left. \begin{matrix} v_{11}^* & v_{21}^* & \dots & v_{t1}^* \\ v_{12}^* & v_{22}^* & \dots & v_{t2}^* \\ \vdots & \vdots & \ddots & \vdots \\ v_{1t}^* & v_{2t}^* & \dots & v_{tt}^* \end{matrix} \right\} \mathbf{v}_1^H$
 $\left. \begin{matrix} v_{12}^* & v_{22}^* & \dots & v_{t2}^* \\ \vdots & \vdots & \ddots & \vdots \\ v_{1t}^* & v_{2t}^* & \dots & v_{tt}^* \end{matrix} \right\} \mathbf{v}_2^H$
 $\left. \begin{matrix} v_{1t}^* & v_{2t}^* & \dots & v_{tt}^* \end{matrix} \right\} \mathbf{v}_t^H$

$$= \mathbf{U} \mathbf{\Sigma} \mathbf{V}^H$$

SVD-based MIMO



define
 $x = V \tilde{x}$

$$y = Hx + n$$

$$y = \underbrace{U \Sigma V^H}_H x + n$$

$$\underbrace{U^H y}_{\tilde{y}} = U^H (U \Sigma V^H x + n)$$

$$\tilde{y} = \underbrace{U^H U}_{I_t} \Sigma V^H x + \underbrace{U^H n}_{\tilde{n}}$$

$$= \Sigma V^H x + \tilde{n}$$

$$\tilde{y} = \Sigma V^H x + \tilde{n}$$

$$= \Sigma \underbrace{V^H V}_{\tilde{x}} \tilde{x} + \tilde{n}$$

$$\tilde{y} = \Sigma \tilde{x} + \tilde{n}$$

$$V^H V = I$$

new norm is $\tilde{n}$

$$\tilde{n} = U^H n$$

$$E |\tilde{n}|^2 = E (|U^H n|^2)$$

↙ Show some steps

$$= E (|n|^2)$$

$$|U^H n|^2 = (U^H n)^H (U^H n)$$
$$= n^H U U^H n$$

$$= n^H n$$

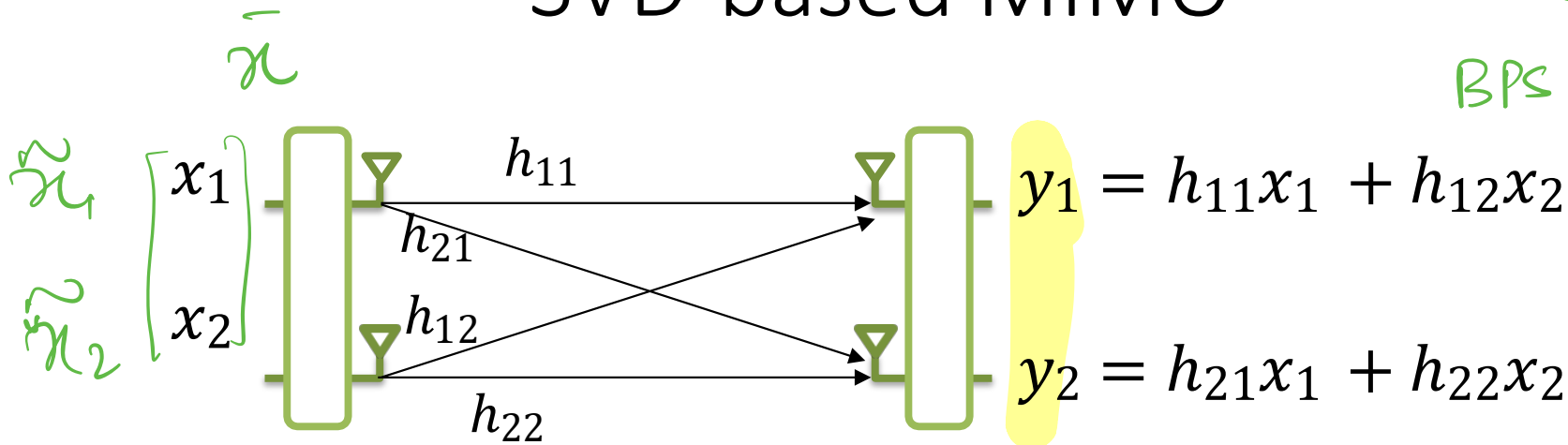
$$= |n|^2$$

$$\boxed{U U^H = I}$$

SVD-based MIMO

$$\tilde{x} = V \hat{x}$$

BPS K symbols are $\tilde{x}$



$$\begin{bmatrix} \tilde{y}_1 \\ \tilde{y}_2 \\ \vdots \\ \tilde{y}_t \end{bmatrix} = \begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_t \end{bmatrix} \begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \\ \vdots \\ \tilde{x}_t \end{bmatrix} + \begin{bmatrix} \tilde{n}_1 \\ \tilde{n}_2 \\ \vdots \\ \tilde{n}_t \end{bmatrix}$$

$$H = U \Sigma V^H$$

NxN MIMO

$$\tilde{y} = \Sigma \tilde{x} + \text{noise}$$

channel singular values

σ_n^2 = noise variance

$$\tilde{y}_1 = \sigma_1 \tilde{x}_1 + \tilde{n}_1$$

$$\tilde{y}_2 = \sigma_2 \tilde{x}_2 + \tilde{n}_2$$

$\vdots$

$$\tilde{y}_t = \sigma_t \tilde{x}_t + \tilde{n}_t$$

$$C_i = \log_2 \left(1 + \frac{P_i \sigma_i^2}{\sigma_n^2} \right)$$

capacity of link 'i'

Total Capacity

$$C = \sum_{i=1}^t \log_2 \left(1 + \frac{P_i \sigma_i^2}{\sigma_n^2} \right)$$

$$\sigma_1 > \sigma_2 > \sigma_3 \dots$$

in 2×2 case only σ_1 & σ_2

Case 1 $\sigma_1 = \sigma_2$

$\Rightarrow$ both data stream
has same SNR

Case 2 $\sigma_1 > \sigma_2$ ($\tilde{n}_1$)

$\Rightarrow$ stream 1 has high
SNR than stream 2

Case when σ_1 is nonzero ($\tilde{n}_2$)
 $\sigma_2 = 0$

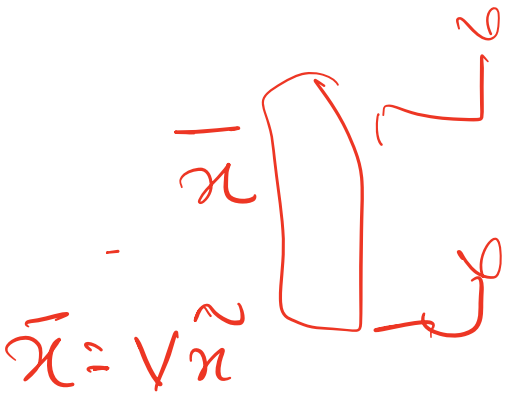
$\Rightarrow$ can only support 1 stream

$\Rightarrow$ can't support 2nd stream

(It was not invertible)

$\Rightarrow$ can still support diversity!

SVD processing

$$\bar{x} = V \tilde{x}$$


H

$$\tilde{y} = U^H y$$


$$\tilde{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

↓
Pre multiply
by U^H

$$\tilde{y} = U^H y$$

load
symbols
in $\tilde{x}$
then
compute
 $\tilde{x}$
 $\tilde{x} = V^H x$

$$y = Hx + n$$

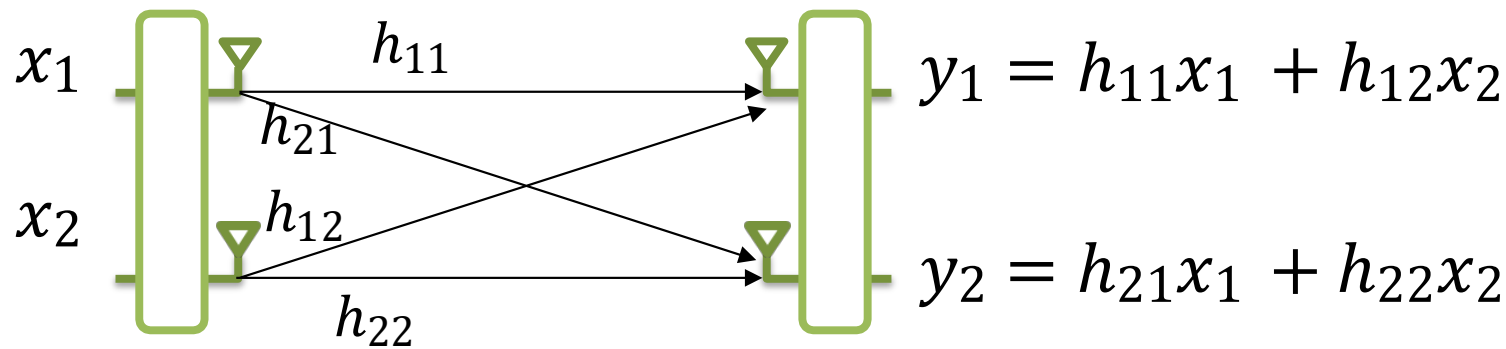
$$\tilde{y} = \sum \tilde{x} + \tilde{n}$$

↑ after $\times$
pre processing

or $\times$ post
processing

⇒ need CSI

SVD-based MIMO: Power allocation?



$$\tilde{y}_1 = \sigma_1 \tilde{x}_1 + \tilde{n}_1$$

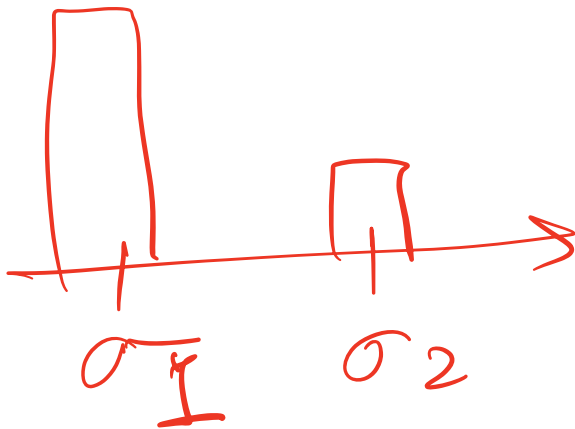
$$\tilde{y}_2 = \sigma_2 \tilde{x}_2 + \tilde{n}_2$$

$$P_1 = E |\tilde{x}_1|^2$$

$$P_2 = E |\tilde{x}_2|^2$$

if equal power

$$P_1 = P_2 = \frac{P}{2}$$



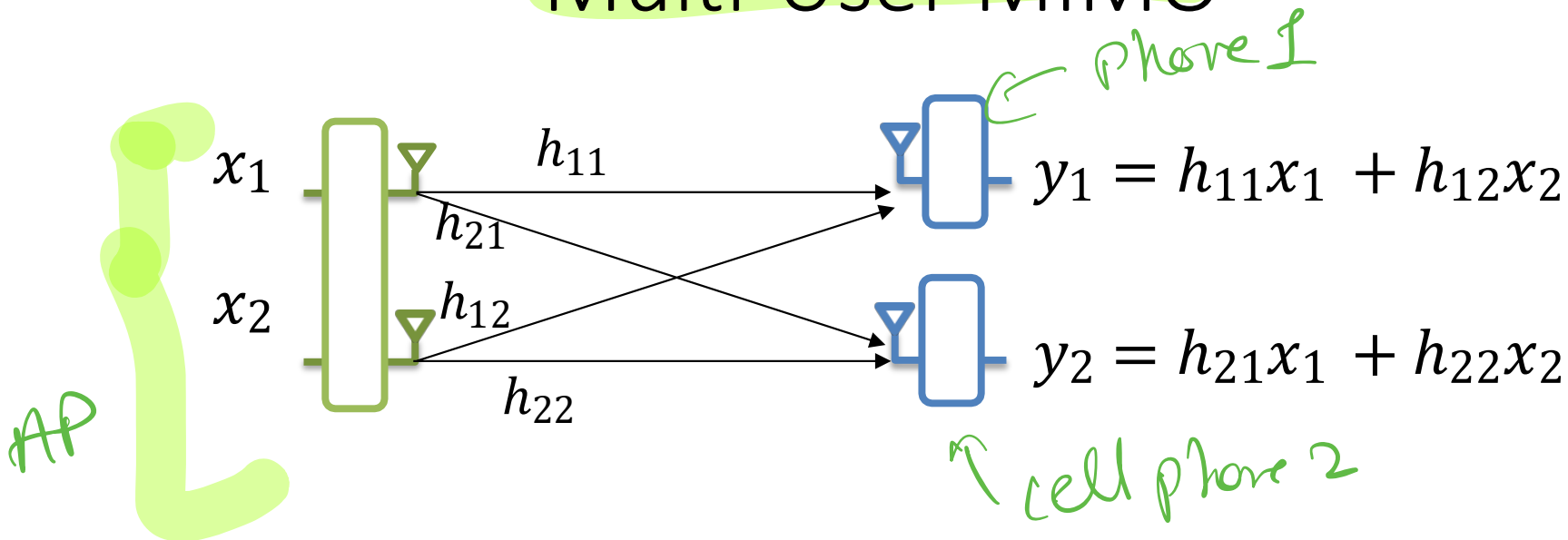
water-filling algo

Total Capacity

maximize

$$C = \sum_{i=1}^t \log_2 \left(1 + \frac{P_i \sigma_i^2}{\sigma_i^2} \right)$$

Multi-User MIMO

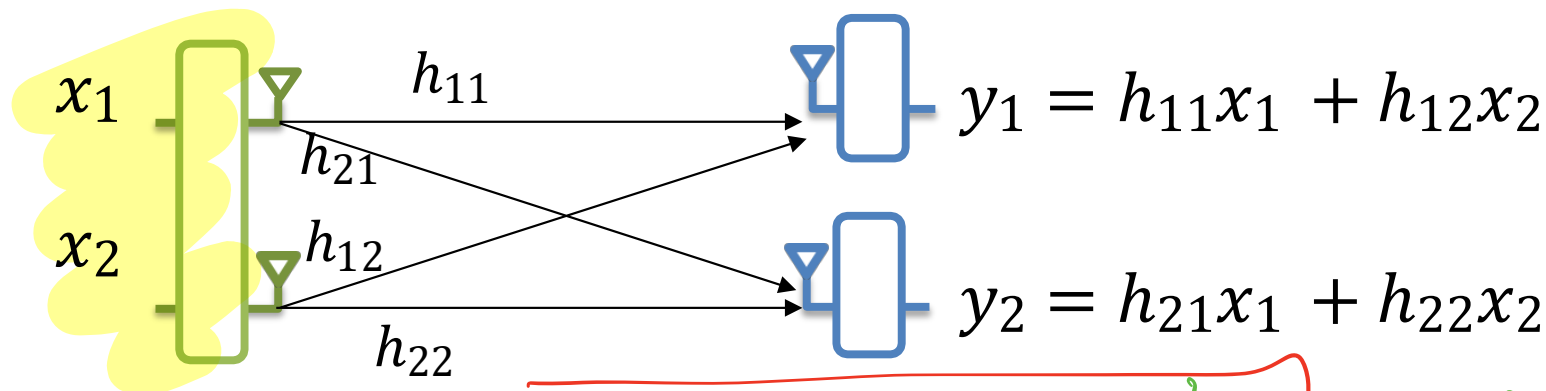


$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$$

$$\tilde{\mathbf{x}} = \mathbf{H}^{-1}\mathbf{y} = \mathbf{H}^{-1}\mathbf{H}\mathbf{x} + \mathbf{H}^{-1}\mathbf{n} = \mathbf{x} + \mathbf{H}^{-1}\mathbf{n}$$

Does not work. Receivers do not have access to each other's signals

Precoding at the Tx



Send: $\tilde{\mathbf{x}} = \mathbf{H}^{-1} \mathbf{x}$ $\tilde{\mathbf{x}} = \mathbf{H}^H (\mathbf{H} \mathbf{H}^H)^{-1} \mathbf{x}$ Assume $\mathbf{H} \mathbf{H}^H = \mathbf{I}$

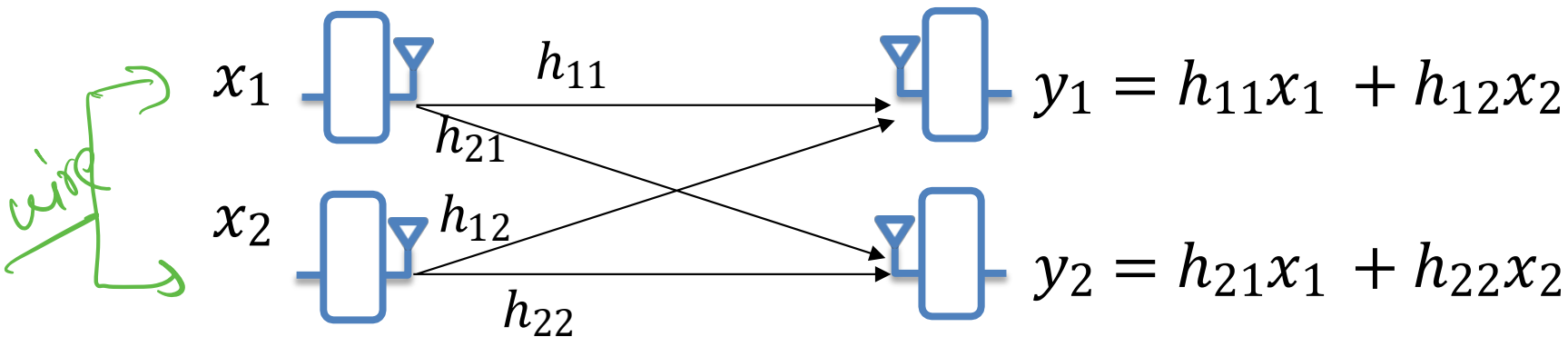
Receive: $\mathbf{y} = \mathbf{H} \tilde{\mathbf{x}} + \mathbf{n} = \mathbf{H} \mathbf{H}^{-1} \mathbf{x} + \mathbf{n} = \mathbf{x} + \mathbf{n}$

$y = H \tilde{x} + n = \cancel{H H^H (H H^H)^{-1}} x + \text{noise}$

Also known as Beamforming or Zero-Forcing

Requires Channel Feedback!

What about distributed transmitters? *distributed MIMO*



Send: $\tilde{\mathbf{x}} = \mathbf{H}^{-1}\mathbf{x}$

Receive: $\mathbf{y} = \mathbf{H}\tilde{\mathbf{x}} + \mathbf{n} = \mathbf{H}\mathbf{H}^{-1}\mathbf{x} + \mathbf{n} = \mathbf{x} + \mathbf{n}$

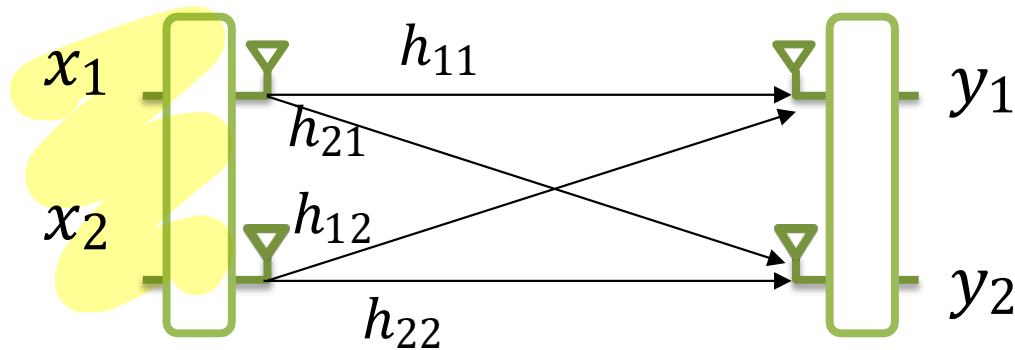
Does not work. Transmitters do not have access to each other's signals

False: what if transmitters are APs connected over Ethernet?

Today's Class

- MIMO Gains
 - Receiver Diversity
 - Transmitter Diversity
 - Space-Time codes (Alamouti Codes)
- MIMO Multiplexing Gains
 - Zero-forcing
 - SVD
 - MU MIMO
- **Clock Synchronization in MIMO**

Clock Synchronization in MIMO

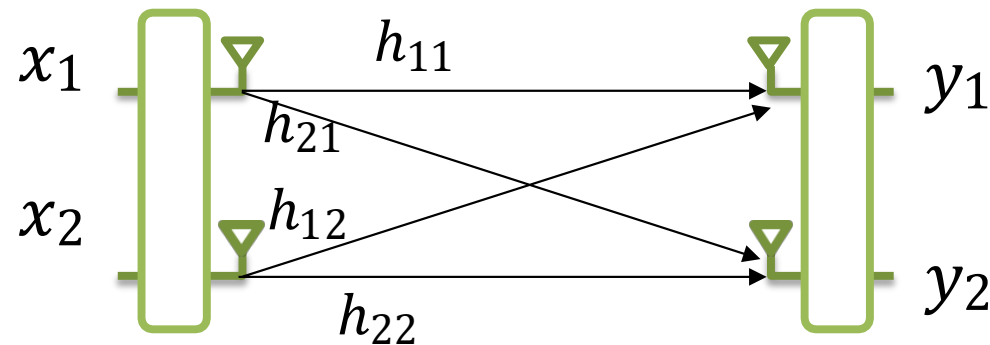


$$y_1 = h_{11}x_1e^{-j2\pi\Delta f_{11}t} + h_{12}x_2e^{-j2\pi\Delta f_{12}t}$$

$$y_2 = h_{21}x_1e^{-j2\pi\Delta f_{21}t} + h_{22}x_2e^{-j2\pi\Delta f_{22}t}$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

Clock Synchronization in MIMO

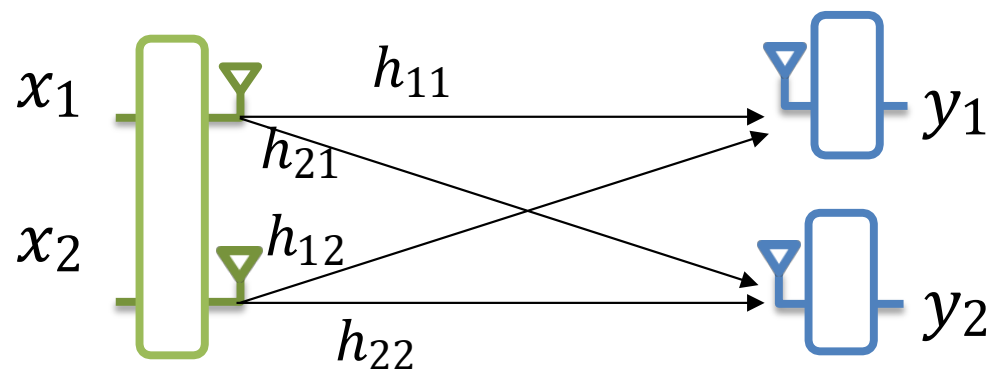


$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} e^{-j2\pi\Delta f_{11}t} & h_{12} e^{-j2\pi\Delta f_{12}t} \\ h_{21} e^{-j2\pi\Delta f_{21}t} & h_{22} e^{-j2\pi\Delta f_{22}t} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

On chip MIMO: $\Delta f_{11} = \Delta f_{21} = \Delta f_{12} = \Delta f_{22} = \Delta f$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} e^{-j2\pi\Delta f t} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

Clock Synchronization in MIMO

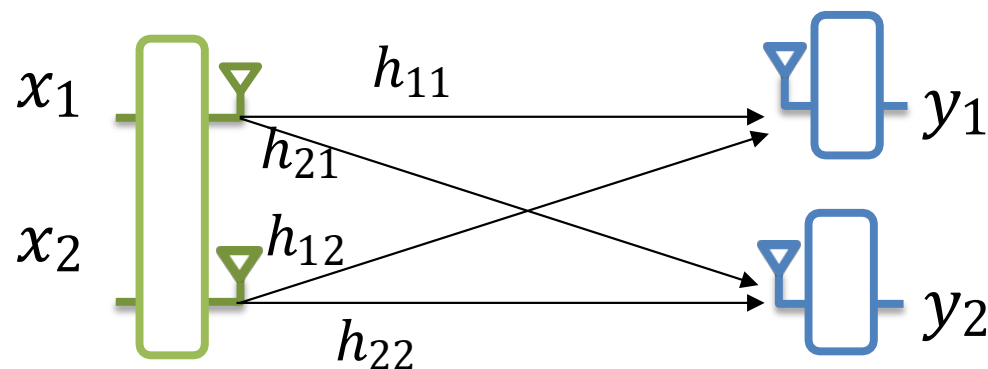


$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} e^{-j2\pi\Delta f_{11}t} & h_{12} e^{-j2\pi\Delta f_{12}t} \\ h_{21} e^{-j2\pi\Delta f_{21}t} & h_{22} e^{-j2\pi\Delta f_{22}t} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

MU-MIMO: $\Delta f_{11} = \Delta f_{12}$ and $\Delta f_{21} = \Delta f_{22}$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} e^{-j2\pi\Delta f_1t} & h_{12} e^{-j2\pi\Delta f_1t} \\ h_{21} e^{-j2\pi\Delta f_2t} & h_{22} e^{-j2\pi\Delta f_2t} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

Clock Synchronization in MIMO

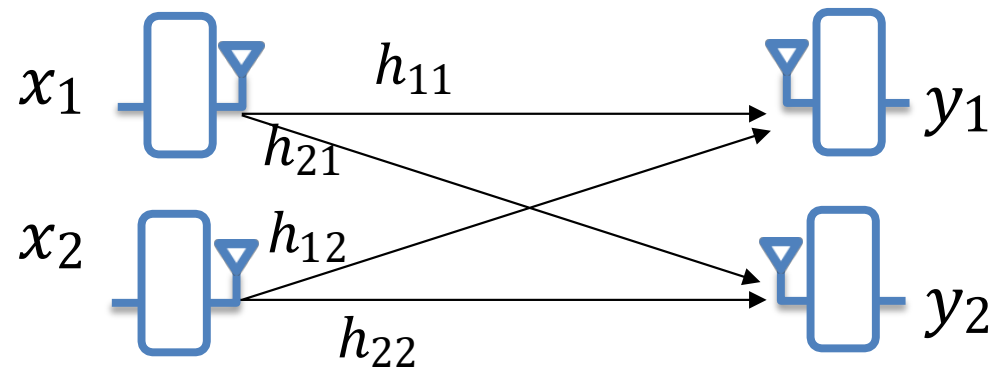


$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} e^{-j2\pi\Delta f_{11}t} & h_{12} e^{-j2\pi\Delta f_{12}t} \\ h_{21} e^{-j2\pi\Delta f_{21}t} & h_{22} e^{-j2\pi\Delta f_{22}t} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

MU-MIMO: $\Delta f_{11} = \Delta f_{12}$ and $\Delta f_{21} = \Delta f_{22}$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} e^{-j2\pi\Delta f_1 t} & 0 \\ 0 & e^{-j2\pi\Delta f_2 t} \end{bmatrix} \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

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$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} e^{-j2\pi\Delta f_{11}t} & h_{12} e^{-j2\pi\Delta f_{12}t} \\ h_{21} e^{-j2\pi\Delta f_{21}t} & h_{22} e^{-j2\pi\Delta f_{22}t} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

Distributed MIMO: $\Delta f_{11} \neq \Delta f_{12} \neq \Delta f_{21} \neq \Delta f_{22}$